

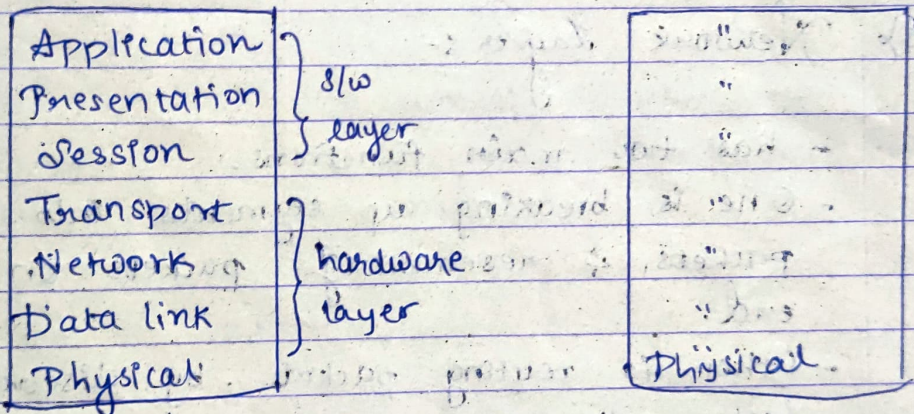
CN

Q.1. Layers of OSI models. (Explain all)

- ⇒
- OSI (Open System Interconnection) is a reference model to explain how diff networking technology work together & interact.
 - Each layer has a specific function for which it is responsible & all the layers work together in correct order to move data around the network.

Sender

Receiver



Physical

1) Application layer

- deals with physical connection betⁿ devices.
- defines hardware elements, like cable, connectors, & network interface cards.
- It is responsible for transmission of the raw data, which is simply series of 0s & 1s, while taking care of bit state control.

2) Data Link Layer:-

- establishes & terminates a connection betⁿ two physically connected nodes on a network.
- It breaks up packets into frames and sends them from source to destination. This layer is composed of two parts:
 - Logical Link Control (LLC) - identifies network protocols, performs error checking & synchronizes frames.
 - Media Access Control (MAC) - uses MAC addresses to connect devices & define perfⁿ permissions to transmit & receive data.

3) Network Layer:-

- has two main functions.
- One is breaking up segments into network packets, & reassembling packets on receiving end.
- Other is routing packets by discovering the best path across a physical network.
- Network layer uses network addresses (IP address) to route packets to a destination node.

4) Transport Layer:-

- takes data transferred in session layer & breaks it into "segments" on transmitting end.
- responsible for resem. reassembling segments on receiving end, turning it back into the data that can be used by session layer.

- carries out flow control, error control, etc.

5) Session Layer.

- creates communicatⁿ channels, called sessions, between devices.
- responsible for opening sessions, ensuring they remain open & functional while data is being transferred, & closing them when communicatⁿ ends.

6) Presentation Layer.

- prepares data for applicatⁿ layer.
- defines how two devices should encode, encrypt, & compress data so it is received correctly on other end.

7) Applicatⁿ Layer.

- used by end-user software such as web browsers & email clients.
- It provides protocol that allow software to send and receive inf. & present meaningful data to users.
- eg of application layer protocols are
 - HyperText Transfer Protocol (HTTP)
 - File Transfer Protocol (FTP)
 - Post Office Protocol (POP)
 - Simple Mail Transfer Protocol (SMTP)
 - Domain Name System (DNS)

Q. OSI reference model. Q. TCP/IP model.

OSI model	TCP/IP model
- stands for Open System Interconnection	- stands for Transmission Control Protocol/Internet Prot.
- has 7 layers	- 4 layers
- low in usage	- It is mostly used
- It is vertically approached	- horizontally approached
- Delivery of package is guaranteed	- not guaranteed
- Replacement of tools and changes can be easily be done	- Replacing tools is not easy as it is in OSI model
- less reliable than TCP/IP model	- more reliable than OSI model

To learn OSI - shortcut

① Vertically ② replacing ③ 7 layers which has
 ④ low usage ⑤ less reliable but ⑥ delivery
 of package is guaranteed

Q. Topology:-

Network Topology:-

is the geometrical representⁿ of relationship betⁿ all links & devices to each other.

Types of Network Topology:-

1) Bus

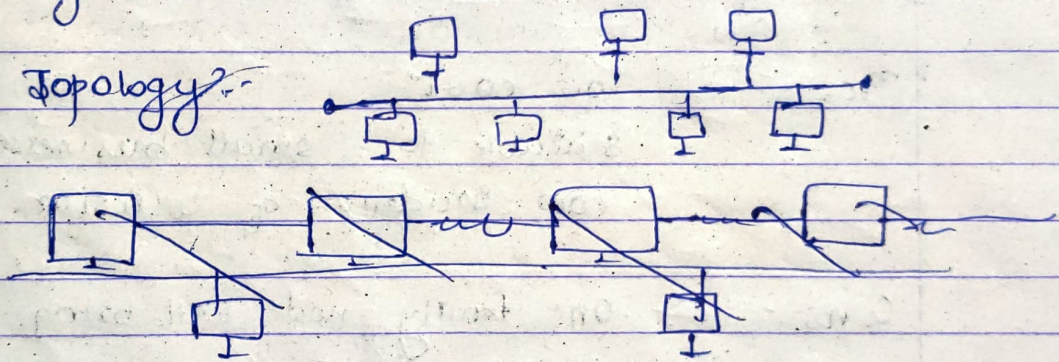
2) Ring

3) Star

4) Mesh

5) Hybrid

1) Bus Topology:-



- Bus topology is a network type where every device is connected to a single cable that runs from one end of network to other
- data is transmitted in one directⁿ

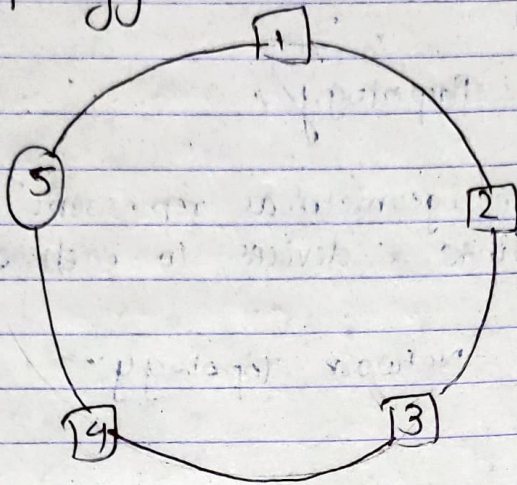
Pros:-

- Easy installatⁿ
- Low cost
- Good for small businesses.

Cons

- Backbone performance is critical
- Data can only travel in one directⁿ

2b Ring Topology



- computers are connected to each other in a circular format.
- Every device in network will have two neighbours.

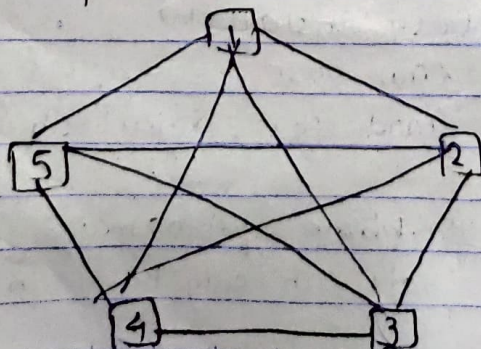
Pros :-

- low cost
- suitable for small businesses
- low incidence of collision

Cons :-

- One faulty node will bring entire network down.
- All data being transferred must pass through each node of network which will make it slower.

3b Mesh Topology



- Every node is connected to each other individually
- No. of connections in a network can be calculated if n is no. of computer devices in network so,

$$\text{no. of links} = \frac{n(n-1)}{2}$$

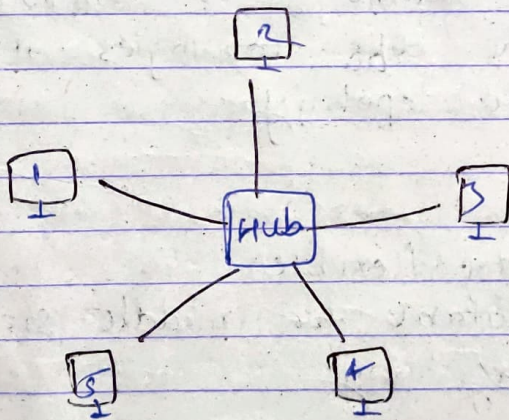
Pros:-

- data transfer is reliable & error free.
- Very secure.
- easy to identify faulty equipment.

Cons:-

- takes long time to set up.
- requires large amt of cable.

4) Star Topology:-



- Every node is connected with one central switch, & indirectly with each other.
- Central network hub is server & other devices are treated as clients.

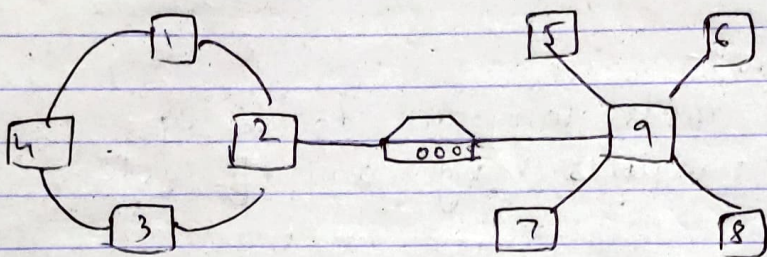
Pros:-

- Simple to setup.
- Simple to manage.

Cons:-

- Dependency on central node
- high setup costs

3) Hybrid Topology



- when a topology is composed of two or more diff. topologies, it is referred as hybrid topology

- Pros:-

- very flexible
- suitable for middle-sized & large org

Cons:-

- req. professional management
- Equipments costs are high

Q3 LAN :- (Local Area Network)

- 1) Group of interconnected nodes over a small area (single room, building).
- 2) Due to short distance, errors & noise are removed.
- 3) Data transfer rate is betⁿ 10-100 mbps.
- 4) Co-axial or CAT-5 cables are used for connection.
- 5) eg:- Comp. lab in school or college.

c) Adv :-

- provides fast data transfer rates
- easy to setup & manage

Disadv:-

- limited geographical coverage
- may experience congestion

MAN (Metropolitan Area Network)

- 1) It is designed to extend over a larger area than LAN's, connecting no. of LAN's form a large network so that resources should be shared.
- 2) Data transfer rate is higher than LAN's.
- 3) extended upto 5-10 kms.
- 4) Adv:-
 - provides high speed connectivity
 - larger geographical area
- 5) Disadv:-
 - can be expensive to setup.
 - may experience congestⁿ.

WAN :- (Wide Area Network).

- 1) These are worldwide network & contain numbers of LANS & MANs.
- 2) Data transfer rate is dependent upon various ISP.

Adv:-

- covers large geog. area
- can connect remote locath

Dis:- can be expensive

- offers slower data transfer rates.

9

Elaborate design issues in CN. RRSSAE

⇒ 1) Reliability:-

Networks & channels & components may be unreliable, resulting in loss of bits while data transfer. So, an imp. design issue is to make sure that inf transferred is not distorted.

2) Scalability:-

Networks are continuously evolving, hence the design should be done so that the network are scalable & it can accomodate some additions & alterations.

3) Addressing:-

At a particular time, innumerable msgs are being transferred betⁿ large no. of comp. So, a naming or addressing system should exist so that each layer can identify the sender & receivers of each message.

4) Error control:-

The layers need to agree upon common error detectⁿ & error correction methods so as to protect data packet while they are transferred.

5) Security:-

There should be an adequate mechanisms to prevent unauthorized access to data through authenticatⁿ & cryptography.

6) Resource allocatⁿ:-

The allocatⁿ should occur so that minimal interference among hosts occurs & there is optimal usage of resources.

Q. Components of Networks:-

1) Router:-

- It is a connecting device which transfers data packets betⁿ diff networks.
- They are used to connect normal node to a wide band network.

2) Repeater:-

- Repeaters receive a signal & regenerate the signal so that it can be transmitted along a long distance.

3) Hubs

- Hubs is a device which is a multiport repeater having several inp/output ports so that input at any port is available at every other port.

4) Switches

A switch receives data from a port & uses diff switching techniques to send data to destination devices.

⇒ Bridge:-

- connects separate ethernet cables

6) Gateway

connects entirely diff. networks that works upon diff. protocols.

7) Network Interface Card (NIC)

component of comp. device which helps to connect to a network.

Diff.

Connection-Oriented services

- 1) designed & developed based on telephone system
- 2) It is used to create an end-to-end connectⁿ betⁿ senders to receiver before transmitt^g data
- 3) creates virtual path betⁿ sender & receiver
- 4) requires authenticatⁿ before transmitt^g data
- 5) No congestⁿ
- 6) TCP is an eg.

Connection-less services

- 1) It is a service based on postal system
- 2) It is used to transfer data packets betⁿ senders to receiver without creat^g any connection.
- 3) does not create virtual path betⁿ S & r.
- 4) does not require authenticatⁿ
- 5) May be congestⁿ
- 6) User Datagram Protocol (UDP), Internet Protocol (IP)

LCARSD.

↑

Q. Explain Various services primitives in CN.

⇒

- ✓ Primitive means operations.
- ✓ service includes set of various primitives.

1) Listen:-

- when a server is ready to accept request of incoming connection, it simply put its primitive into action.
- Its simply waiting for incoming connectⁿ req.

2) CONNECT:-

- used to connect server simply by creating or establishing connectⁿ with waiting peer.

3) ACCEPT:-

- simply accepts incoming connectⁿ from peer.

4) RECEIVE:-

- These primitive afterwards block the server.
- Receive simply waits for incoming msg.

5) SEND:-

- It is put into action by the client to transmit its req. that is followed by putting receive primitive into action to get the reply.
- simply sends msg to peer.

6. Disconnect:-

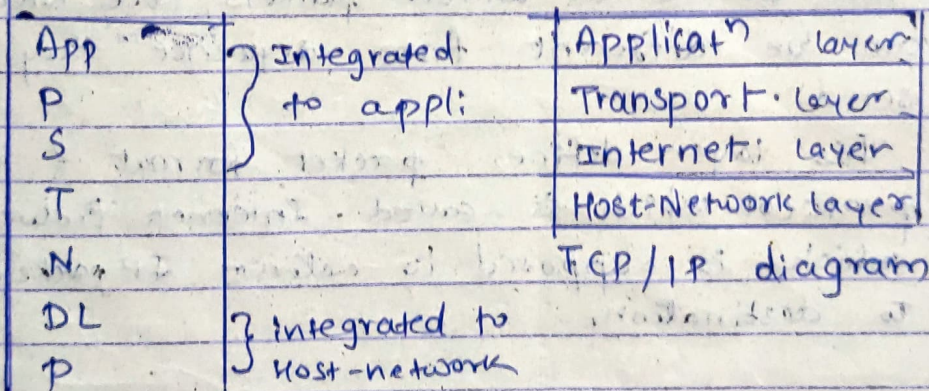
- simply used to terminate or end connectn after which no one will be able to send any of the message.

Q11. Layer Diagram of TCP/IP protocol suite & explain.

⇒ It is a concise version of OSI model.

It contains four to five layers, unlike seven layers in OSI model.

⇒ The main work of TCP/IP is to transfer data of a comp from one device to another.



OSI model

Fig. No. 1. TCP/IP model.

⇒ Applicatⁿ - works like an interface betⁿ applicatⁿ program & interacts with them.

- In TCP/IP, does not have session & presentatⁿ layer because they are not that relevant to TCP/IP.
- It is on top of transport layer.
- The protocols related to this layer are high level protocols such as FTP, SMTP, HTTP, HTTPS, SSH, etc.

2) Transport:-

It functions exactly same as OSI model's Transport layer.

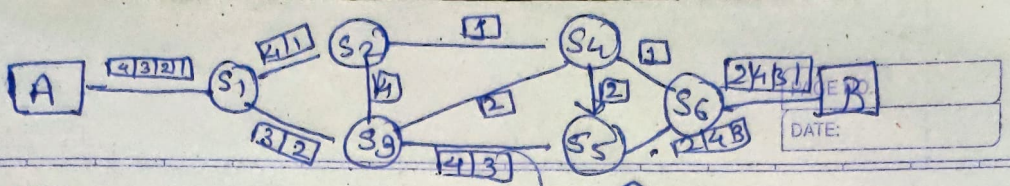
- The end-to-end protocol are used here like TCP & UDP.
- TCP is reliable connectⁿ oriented protocol.
- ensures delivery of each bit stream from one to another machine error free.
- TCP control → flow control

3) Internet:-

- holds whole architecture together.
- task is to allow host device to insert packets into the network & make them travel independently.
- The orders in which packets are received can be different from sequence they are send.
- This layer defines packet format & a protocol which is called Internet Protocol (IP) which is supposed to deliver IP packets to destination.

4) Host to Network layer:-

- lowest layer of TCP.
- connects device using some protocol so that it can send IP packets over a network.
- This protocol form host to host & network to network.



Q.12

Packet Switching

- 1) In packet switching, the messages are sent in one go, but they are divided into smaller packets & transferred independently.
- 2) Each msg. is divided into packets are given a sequence number for packet identification.
- 3) Every packet contains info in their header like source, destination address & seq. no.
- 4) These packets travel around the network taking shortest path as possible.
- 5) All the packets are reassembled in receiving end in correct order. In case of any missing or corruption, packets are retransmitted.
- 6) And an acknowledgement is sent after successful delivery of msg.
- 7) It follows two approaches:-

- 1) Virtual packet switch
- 2) Datagram p.s.

Pros:-

- very cost-effective
- efficient.

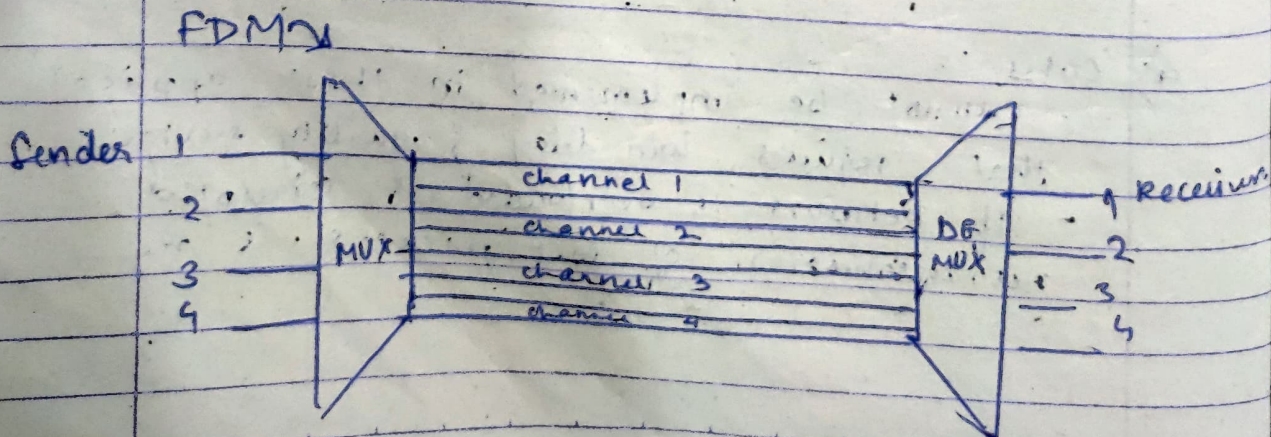
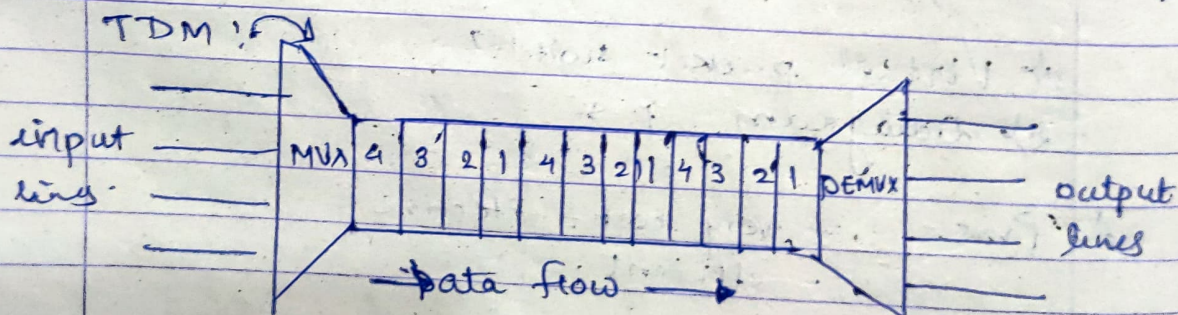
Cons:-

- cannot be implemented in those applications that require low delay & quality service.
- If network is overloaded & corrupted, retransmission of lost packets is req.

Q

Distinguish betⁿ TDM & FDM.
suitable diagram,

TDM	FDM
1) stands for Time division multiplexing.	1) stands for Frequency Div multiplexing.
2) works with digital signals as well as analog signals.	2) works with only analog signals.
3) has low conflict.	3) has high conflict.
4) wiring on chip of TDM is simple.	4) It is complex.
5) efficient.	5) Inefficient.
6) time sharing takes place.	6) Freq. sharing takes place.



Q. Circuit Switching

1) In c.s, 3 phases:-

i) connectⁿ establishment

ii) Data Transfer

iii) Connectⁿ released

2) Each data units knows entire path address which is provided by source

3) more reliable

4) Wastage of resource is more in c.s.

5) Transmission of data is done by source

6) Call setup is req.

7) req. simple protocols for delivery

Packet Switching

1) In p.s, directly data transfer takes place

2) Each data unit just knows final destinatⁿ address, intermediate path is decided by routers.

3) less reliable

4) less wastage of resource compared to c.s.

5) Trans. of data is done not only by sources but also by intermediate routers

6) No call setup.

7) requires complex protocol for delivery

Q Guided mode Transmission	Unguided Mode Transmission
1) In guided media, signal energy communicates via wires.	1) In unguided media, signal energy communicates through air.
2) It is generally preferred when we want to execute direct communication.	2) It is generally preferred for radio broadcasts in all directions.
3) signals are in state of current & voltage.	3) are in state of electromagnetic waves.
4) cost-effective	4) expensive
5) eg:- Twisted pair wires, coaxial cables, optical fiber cables.	5) eg:- microwave or radio links, infrared light.
6) It is unable to pass through walls.	6) can pass through walls.

To learn → Guided mode

- • signal energy communicates wires, & direct communication to the types of wires & current & voltage, which is cost-effective & unable to pass walls.



Q. Optical fibre transmission.

Types of propagation in data communication

1) Line of sight (LOS) propagation.

• Among modes of prop., the LOS is the one which we commonly notice.

• As the name implies, the wave travels a min. distance of sight, which means it travels to the distance of sight up to which naked eyes can see.

• LOS will not be smooth if there occurs any obstacle in its transmission path.

2) Grounded wave

• follows contour of earth surface wave is called direct wave

• The wave sometimes bends with due to earth's magnetic field & get reflected to receiver, such wave can be termed as reflected wave

• The wave when propagated with earth's atmosphere is known as ground wave.

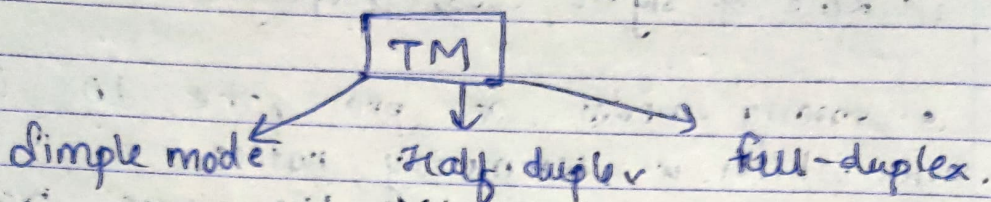
3) Sky wave prop.

• It is preferred when wave has to travel a longer distance.

• A travel of wave from transmitter to the ionosphere & from there to receiver on earth is known as sky wave.

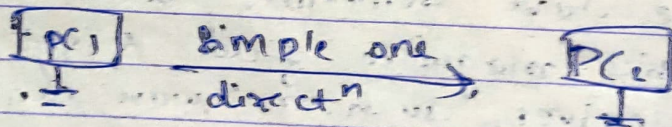
Q. Transmission Modes:-

* Transferring data betⁿ two devices, also known as communicatⁿ mode



1) Simple :-

- communication is unidirectional.
- Only one of two devices on a link can transmit, the other can only receive.
- ex:- keyboard & traditional monitors.
Keyboard can only introduce input
monitors can only give o/p

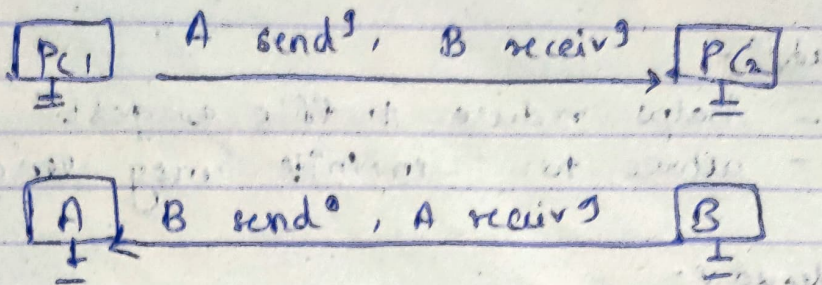


2) Half duplex:-

1) Each station can both transmit and receive, but not at the same time.

2) when one is sending, the other can only receive & vice versa

3) eg Walkie-talkie. In which msg is sent at one time & msg are sent in both directⁿ.

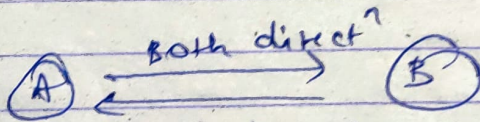


3) Full duplex mode :-

- 1) In FDM, both station can transmit and receive simultaneously.
- 2) Full duplex is used when communication in both direction is req. all time.

3) Ex:-

Telephone network in which there is communication betⁿ two persons by a telephone line through which both can talk and listen at same time.



Message Switching

- - End users communicate by send^g & receiv^g msg that included the entire data to be shared.
- Msg is smallest individual units.
 - Also, sender & receiver are not directly connect^d.
 - There are no. of intermediate nodes that can transfer data & ensure that msg reaches its destination.
 - called as hop-by-hop systems.

adv:-

- helps reduce traffic congestⁿ.
- allows for infinite msg lengths.

disadv:-

- cannot be used for real-time applicatⁿ as stor^g msg^s can cause delay.
- does not create dedicated path

Q. Circuit Switch^g techniques:-

⇒ It is divided into 3 pieces and has dedicated path est. betⁿ sender & receiver providing a guaranteed data rate.

Phases:-

1) Circuit Est:

A dedicated circuit betⁿ source & destinatⁿ is est. with help of numerous intermediate switching centres.

2) Data Transfer:-

The transfer of data & voice signals are possible betⁿ source and destinatⁿ after est. of circuit.

3) Circuit disconnect:-

The disconnect happens when one of user initiates to disconnect. When dis. takes place, all intermediate links betⁿ sender & receiver are removed.

Adv:-

- suitable for continuous transmissⁿ of data.
- rate of communicatⁿ is steady.

Disadv:-

- establishment time is high.
- as there is est. of dedicated connectⁿ, the transmission of any other data is challenging.